

Tetrahedral sums of Hermitian matrices

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Abstract

TODO: write an abstract

1 Introduction

TC: This is the main document for this project. Previous drafts are located in the folder titled “old” and will not necessarily compile properly.

2 Tetrahedra & Hermitian matrices

2.1 Definition

2.2 Symmetries

2.3 Special cases

3 Non-linear inequalities

Lemma 1. *Let $A, B, C \geq 0$ be positive semidefinite, then*

$$\sum_j \lambda_j(A+B)^2 + \sum_j \lambda_j(B+C)^2 \leq \sum_j \lambda_j(B)^2 + \sum_j \lambda_j(A+B+C)^2. \quad (1)$$

Proof. Note that $\sum_j \lambda_j(X)^2 = \text{Tr}[X^2]$ for any Hermitian operator X . Therefore,

$$\begin{aligned} & \text{Tr}[(B)^2] + \text{Tr}[(A+B+C)^2] - \text{Tr}[(A+B)^2] - \text{Tr}[(B+C)^2] \\ &= (\text{Tr}[B^2]) + (\text{Tr}[A^2] + \text{Tr}[B^2] + \text{Tr}[C^2] + 2\text{Tr}[AB] + 2\text{Tr}[BC] + 2\text{Tr}[AC]) \\ & \quad - (\text{Tr}[A^2] + \text{Tr}[B^2] + 2\text{Tr}[AB]) - (\text{Tr}[B^2] + \text{Tr}[C^2] + 2\text{Tr}[BC]) \\ &= 2\text{Tr}[AC]. \end{aligned} \quad (2)$$

Since $A, C \geq 0$ are both positive semidefinite, we also have $\text{Tr}[AC] \geq 0$ which proves the claim. $\square$

3.1 Proof via spectral embedding

3.2 Proof via representation theory

4 Conclusion

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5 [Research Notes]

5.1 Horn’s problem & generalizations

- Horn’s original paper where they conjectured the form of a necessary and sufficient set of inequalities via an inductive scheme for increasing dimension [Hor62]
- An excellent expository overview of the big picture by Knutson [Knu00]
- An excellent introduction to these ideas is the recent masters thesis on various aspects of additive and multiplicative Horn problems in connection to Vinnikov curves and the Lax conjecture due to Hess and supervised by S. Kuhlmann and M. Infusino [Hes20]
- Quantum-info style proof of Littlewood-Richardson coefficient connection due to Christandl [Chr08]
- Coquereaux and Zuber explicitly outline the link between the orbital measures (i.e. the induced distributions in the sense of Klyachko’s random walk approach) and the magnitude of the stretched Littlewood-Richardson coefficients [CZ18]. See also the papers [CMZ20; CMZ19; Zub18].
- A recent article by McSwiggen is an excellent introduction to the Harish-Chandra integral [McS21]. The proof of the Harish-Chandra formula provide by McSwiggen using the heat equation methods from Fegan’s 1978 paper [Feg78] seems very promising for deriving a connection to the character formula for the norm of the recoupling operator!
- Speyer [Spe05]
- Generalizations of Horn’s problem to the product of unitary matrices [AW97; Bel08; BK16]
- Bhatia [Bha01]
- History account offered by Daftuar and Hayden [DH05], and technical review article by Fulton [Ful00].
- Berline, Vergne, and Walter a pedagogical account of the triple role played by Horn’s inequalities in characterizing the existence of invariant vectors in tensor products, the eigenvalues of sums of Hermitian matrices, and intersections of Schubert varieties [BVW18]
- Fulton provides a comprehensive treatment of the connection between representation theory of $GL(d)$ and the geometry and (co)homology of the complex Grassmannians [Ful97]
- Prakash Belkale has contributed greatly to the theory surrounding Horn-type problems: for example, in [Bel04], Belkale provides a *geometric* proof of the equivalence between (i) the structure coefficients of the representation theory of $GL(d)$ (i.e. the LR-coefficients $c_{\mu\nu}^\lambda$), and (ii) the structure coefficients of the cohomology ring of the Grassmannian $Gr(d, n)$ of d -dimensional subspaces in $\mathbb{C}^d$. This proof is now complementary to the previously known algebraic proofs which rely on knowledge of the algebra of symmetric functions. This geometric approach taken by Belkale, which emphasizes the tangent spaces of the involved Schubert varieties, is later used by Belkale to provide a geometric proof of the saturation conjecture [Bel02].

5.2 Thompson’s conjecture

- Original (indirect) proof through the lense of random walks Klyachko [Kly00]
- The proof of Thompson’s conjecture in the style of Alekseev, Meinrenken, and Woodward is the “linearization paper” due to [AMW01].
- See also the related article due to Lamb which summarizes the ideas of [AMW01] while also providing more technical background [Lam05]
- Lu and Evens extend the proof technique of [AMW01] to a larger class of Lie groups [LE03]

- Alekseev, Podkopaeva, and Szenes [APS17a]
- Alekseev, Podkopaeva, and Szenes [APS17b]
- Alekseev and Meinrenken [AM07]
- Deforming the symplectic volume by the s parameter Alekseev et al. [Ale+18]
- **TODO:** track down pdfs for the original triple of papers by Thompson and Therianos titled “On the singular values of matrix products” I, II, and III from 1973
- **TODO:** track down pdfs for the original triple of papers by Thompson and Freede titled “On the eigenvalues of sums of Hermitian matrices” I, II, and III from 1973
- **TODO:** track down a pdf for Thompson’s 1988 “Matrix Spectral Inequalities” referenced by Klyachko
- Thompson and Therianos [TT73] present a number of interesting counter-examples to the question of whether the inequalities pertaining to the spectra of general matrices can be deduced from looking at the spectra of diagonal matrices (both in the additive and multiplicative cases).

5.3 $6j$ symbols

- For a comprehensive reference textbook for the $6j$ coefficients, see the textbook by Varshalovich, Moskalev, and Khersonskii [VMK88], specifically Chapter 9.
- A great summary by Roberts [Rob02]; also the original result for the asymptotics of the $6j$ symbols due to Roberts [Rob99]
- The masters essay/thesis of Backens (**TODO: check if the pdf Matthias sent to me is the final version**) [Bac10]
- The spectral quantum marginal problem and the $6j$ symbol for the symmetric group Christandl, Şahinoğlu, and Walter [CSW18]
- Usage of Wigner’s $6j$ symbol to compute asymptotic error rates for the promise problem of distinguishing between two possible “change points” for a sequence of unknown pure states to transition from one unknown type to another [AH11]
- See also Coquereaux’s later paper which promotes the alternative, albeit equivalent, usage of the integer-valued gon and tet functions in place of the rational-valued theta (triangular) and irrational-valued $6j$ symbols [Coq23]. Most relevant for this project is [Coq23, Section IV D] which discusses the apparent lack of substantial generalization from $SU(2)$ to $SU(N)$ (“*apart from very few special cases (low dimension of irreps and no multiplicity) worked out by physicists at the time of the eightfold way, we are not aware of any published general result*”).
- The recent paper by Alekseev, Morozov, and Sleptsov (different Alekseev!) covers the case of some multiplicity free formulas for the quantum $6j$ symbols for $U_q(\mathfrak{sl}_N)$ [AMS20]
- Keppeler, Plätzer, and Sjodahl recently calculated $SU(d)$ $6j$ symbols with adjoint representations [KPS24] and a bit earlier [Alc+23] calculated $SU(d)$ $6j$ symbols with at least two fundamental representations [Alc+23].
- Representation theoretic proof of the Regge symmetries due to Boalch [Boa07]
- Asymptotics of quantum $6j$ symbols Taylor and Woodward [TW06]
- The integral formula for the square of the $SU(2)$ $6j$ symbol can be performed by an appropriate change of variables [FL03]. See also other papers by Laurent Freidel and collaborators that are relevant [FK00; FNR07; FKL10].

- Alcock-Zeilinger consider the case of $6j$ symbols for $SU(N)$ for wherein two opposing edges are labelled by fundamental irreps of $SU(N)$ so that all vertices are multiplicity one (or zero) [Alc18] TC: They also freely use the birdtrack notation which I enjoy.
- Only the tetrahedral symmetry of the $U_q(SL(N))$ $6j$ -symbol is known, the counterpart of the Regge symmetry is still unknown, although Morozov and Sleptsov conjectures the generalized Regge symmetry may be related to the eigenvalue conjecture, just as it is in the case when $N = 2$ [MS18].
- The graphical notion, specifically the use of an additional edge for multiplicity spaces can has been used for a long time, as early as [AB68; Mou84].
- The original Ponzano-Regge paper [PR68]
- Gurau derives the Ponzano-Regge asymptotic formula directly using Stirling’s approximation applied to Racah’s single-sum formula [Gur08]
- Bartlett and Nzaganya work with so-called *coherent loop states* or *Bohr-Sommerfeld states* to develop some semiclassical results concerning the inner products between such states (see also the talk on Youtube) [BN23]
- Aquilanti et al. provides an excellent introduction to subject of classical $6j$ symbols, their history, diagrammatics, presentations and asymptotics [Aqu+12]
- Marzuoli and Rasetti connects quantum computing models to the recoupling theory of $SU(2)$ angular momentum and also considers the semiclassical limit [MR05]
- Artamonov derives an explicit formula for the $d = 3$ case [Art23], and later to arbitrary d [Art24].
- There is a series of papers involving Alexey Sleptsov, starting with a computation of quantum $6j$ symbols for generic dimension d but with symmetric representations in terms of certain hypergeometric polynomials [MMS17], then with a discussion of the eigenvalue hypothesis and its potential connection to generalized Regge symmetries [MS18], then with a calculation of all multiplicity free quantum $6j$ symbols for arbitrary dimension [AMS20], and then finally with the discovery of novel “tug-the-hook” symmetries for quantum $6j$ symbols for all d [LS23].

5.4 Minkowski’s theorem and tetrahedron geometry

- Connecting group-valued moment maps to spherical, flat, and hyperbolic tetrahedron via Minkowski’s theorem from the reconstruction of flat tetrahedron Haggard, Han, and Riello. See also the comments in [HHR16, Section 11] on the utility of *quasi-poisson* (controlled failure of the Jacobi identity) manifolds and group-valued moment maps.
- Foundational results concerning the relationships between the edge lengths, face areas, dihedral and solid angles of a tetrahedron by Richardson [Ric02]
- Bianchi, Dona, and Speziale uses Minkowski’s theorem to parametrize convex polyhedra in $\mathbb{R}^3$ with specified face areas and the associated Kapovich-Millson phase space (i.e., the *space of shapes of polyhedra with fixed areas* and then considers its connection to loop quantum gravity via the usual Guillemin-Sternberg quantization commutes with reduction philosophy which associates invariants of the quantization of the unreduced phase space to the reduced phase space [BDS11] (notice the connection with [HHR16]).
- Regarding an algorithmic construction for the polyhedra appearing in Minkowski’s theorem and their complexity, see this post: [constructive-algorithm-for-minkowskis-theorem](#)
- Charles quantization of polygon spaces, tetrahedron and semiclassical limits [Cha10]

5.5 Quantization, Poisson geometry & quantum groups

- A popular textbook on quantum groups with an emphasis on the connection to Poisson manifolds, Lie bialgebras and their quantizations by Chari and Pressley [CP95]
- Course notes on Poisson geometry with an eye toward quantization by Ciccoli and Witkowski [CW06]
- The PhD thesis of Esposito (from the maths department here in Copenhagen) serves as a introduction to the world of Poisson momentum maps and the notion of a “quantum” momentum map as a deformation of the ordinary “classical” momentum map (where deformation is in the sense of deformation quantization) [Esp12].
- Alekseev [Ale97]
- Papers contained results pertaining to the generalization of Duistermaat-Heckman measures for Hamiltonian group actions with group-valued moment maps by Alekseev, Meinrenken, and Woodward [AMW99; AMW00]
- Kirillov and Reshetikhin [KR90]
- Octahedron Recurrence/Rule Robbins and Rumsey Jr [RR86] applied to the “Hive” ring Knutson, Tao, and Woodward [KTW03], and also Henriques and Kamnitzer [HK06]
- Henriques and Kamnitzer [HK05]
- A reference which contains the formula for the quantum dimension of a representation of $U_q(\mathfrak{sl}(d))$ labelled by λ , thanks Rinat! [Ais97]. See also [AM98] which clarifies the relation between various parameters. Apparently, the x, v, s parameters are apparently natural parameters to consider in the context of invariants of knots and links. The substitution mentioned in [AM98] is the following

$$s = e^{h/2}, \quad v = e^{-dh/2} = s^{-d}, \quad x = e^{-h/2d} = s^{-\frac{1}{d}}, \quad q = e^h = s^2.$$

This is consistent with [Res87] in the sense that

$$[k] := \frac{s^{+k} - s^{-k}}{s^{+1} - s^{-1}} = \frac{q^{+k/2} - q^{-k/2}}{q^{+1/2} - q^{-1/2}}. \quad (3)$$

- Reshetikhin develops the diagrammatics of the category of representations of $U_q(\mathfrak{g})$ motivated by connections to the developing theory of polynomial invariants of knots and links [Res87].
- Moak investigates a q -Stirling’s formula for the logarithm of the q -factorial [Moa84].

5.6 Trace inequalities

- The general subject of multivariate trace inequalities involving exponentials of sums of Hermitian and products of their exponentials is, of course, intimately related. See for instance the paper by Sutter, Berta, and Tomamichel [SBT17].
- Also the article [FT14] by Forrester and Thompson which describes the connection between Thompson’s inequality and hyperbolic geometry quite explicitly.
- Friedland and Gaubert considers submodular functions associated to operator monotones evaluated on different principal submatrices of a single Hermitian matrix [FG13]. While it is not clear to be yet whether or not one can relate partial sums of matrices to principal submatrices, [FG13] references a paper by Bouhtou, Gaubert, and Sagnol [BGS08] which proves the following family of inequalities for all positive semidefinite $A, B, C \geq 0$ (and $p \in [0, 1]$):

$$\sum_j \lambda_j (A + B)^p + \sum_j \lambda_j (B + C)^p \geq \sum_j \lambda_j (B)^p + \sum_j \lambda_j (A + B + C)^p. \quad (4)$$

- For multivariate inequalities based on idempotents of the symmetric group algebra see Huber’s paper [Hub20].
- For spectral inequalities derived using semidefinite programming, see Huber and Wyderka’s paper [HW22].

5.7 Uncategorized

- Elementary introduction to toy models of three-dimensional quantum space [Bar03]
- Regge and Williams [RW00] a relatively recent review of discrete approaches to gravity. The essential idea, in contrast to the differential geometry of general relativity, is to model curvature discretely instead of continuously. In particular, one can think of a triangulation of the surface of a sphere; the curvature of the sphere is encoded in the triangulation when a configuration of three (or more) triangles meet at a common vertex with sum of interior angles deviating from π . It is in this sense that Regge and Williams refers to these types of models as “piecewise-flat”. Contained is an overview of various attempts to generalize discrete theories of three-dimensional gravity to four dimensions [RW00, Section III].